

Dry Run Exercises - Solutions

Unit 2 Variables, Assignment and Expressions

Complete the dry run table for each fragment of code given in the questions below. **Give** the value(s) printed to the screen when each is executed.

Follow the module dry run style guide in your answers.

1.

```
int x = 22;           // L1
int y = 16;           // L2
x = y;                // L3
System.out.println(x + "." + y); // L4
```

Complete the dry run table (you may not need all rows):

LINE	x	y	printed
L1	22		
L2		16	
L3	16		
L4			16.16

Write EXACTLY what is printed here

16.16

For what is printed you should have written in the box exactly the above (and nothing else).

2.

```
int num1 = 1;           // L1
int num2 = 3;           // L2
num1 = num2 + 1;        // L3
System.out.println(num1 + "." + num2); // L4
```

Complete the dry run table (you may not need all rows):

LINE	num1	num2	printed
L1	1		
L2		3	
L3	4		
L4			4.3

Write EXACTLY what is printed here

4.3

For what is printed you should have written in the box exactly the above (and nothing else).

3.

```
int a = 2;              // L1
int b = 4;              // L2
b = a + a + a;          // L3
System.out.println(a);  // L4
System.out.println(b);  // L5
```

Complete the dry run table (you may not need all rows):

LINE	a	b	printed
L1	2		
L2		4	
L3		6	
L4			2
L5			6

Write EXACTLY what is printed here

2
6

For what is printed you should have written in the box exactly the above (and nothing else).
As there are two println statements they print on different lines.

4.

```
int a = 4;           // L1
int b = 2;           // L2
b = a * a;           // L3
System.out.print(a); // L4
System.out.print(b); // L5
```

Complete the dry run table (you may not need all rows):

LINE	a	b	printed
L1	4		
L2		2	
L3		16	
L4			4
			16

Write EXACTLY what is printed here

416

For what is printed you should have written in the box exactly the above (and nothing else). As there are two print (note not println) statements they print on the same line as they do not insert a new line after printing the string.

5.

```
String colour1 = "BLUE";           // L1
String colour2 = "RED";             // L2

System.out.println(colour2 + colour1); // L3
```

Complete the dry run table (you may not need all rows):

LINE	colour1	colour2	printed
L1	"BLUE"		
L2		"RED"	
L3			REDBLUE

Write EXACTLY what is printed here

REDBLUE

For what is printed you should have written in the box exactly the above (and nothing else). There are no spaces between the two strings printed as no spaces were included in the string to print.

6.

```
String colour1 = "BLUE";           // L1
String colour2 = "RED";            // L2
String temp = "";                  // L3

temp = colour1;                    // L4
colour1 = colour2;                 // L5
colour2 = temp;                    // L6

System.out.println(colour1);        // L7
System.out.println(colour2);        // L8
```

Complete the dry run table (you may not need all rows):

LINE	colour1	colour2	temp	printed
L1	"BLUE"			
L2		"RED"		
L3			" "	
L4			"BLUE"	
L5	"RED"			
L6		"BLUE"		
L7				RED
L8				BLUE

Write EXACTLY what is printed here

RED
BLUE

For what is printed you should have written in the box exactly the above (and nothing else). Note an empty string can be shown as "". The code swapped the values in the variables over.

7.

```
int y = 7;          // L1
int x = 8;          // L2
```

```
x = y;              // L3
y = x;              // L4
```

```
System.out.println(x);    // L5
System.out.println(y);    // L6
```

Complete the dry run table (you may not need all rows):

LINE	y	x	printed
L1	7		
L2		8	
L3		7	
L4	7		
L5			7
L6			7

Write EXACTLY what is printed here

7 7

For what is printed you should have written in the box exactly the above (and nothing else). Remember assignments happen one after another in the order given, not at the same time.

8.

```
int x = 1;           // L1
```

```
x = 4;              // L2
```

```
x = 6;              // L3
```

```
System.out.println(x);      // L4
```

Complete the dry run table (you may not need all rows):

LINE	x	printed
L1	1	
L2	4	
L3	6	
L4		6

Write EXACTLY what is printed here

6

For what is printed you should have written in the box exactly the above (and nothing else). Only one value can be stored in a variable - the last value put there.

9.

```
int x = 7;          // L1
int y = 8;          // L2
int z = 9;          // L3
```

```
x = y;              // L4
y = z;              // L5
z = x;              // L6
```

```
System.out.println(x + y + z);          // L7
```

Complete the dry run table (you may not need all rows):

LINE	x	y	z	printed
L1	7			
L2		8		
L3			9	
L4	8			
L5		9		
L6			8	
L7				25

Write EXACTLY what is printed here

25

For what is printed you should have written in the box exactly the above (and nothing else).

Variables hold only the last value.

The + here is interpreted as addition not string concatenation as the arguments are numbers. Note the difference between

```
System.out.println(x + y + z);
```

and

```
System.out.println(x + " " + y + " " + z);
```

The latter does string concatenation. as each pair of operands to + includes a String, so addition no longer makes any sense.

10.

```
char a = 'a';           // L1
char b = 'b';           // L2
char c = 'c';           // L3

a = b;                  // L4
System.out.println(a + ", " + c);    // L5

b = c;                  // L6
c = a;                  // L7

System.out.println(b + ", " + c);    // L8
```

Complete the dry run table (you may not need all rows):

LINE	a	b	c	printed
L1	'a'			
L2		'b'		
L3			'c'	
L4	'b'			
L5				b,c
L6		'c'		
L7			'b'	
				c,b

Write EXACTLY what is printed here

```
b,c
c, b
```

For what is printed you should have written in the box exactly the above (and nothing else).

If you missed the space or the comma in the second line printed (or put a space in the first) then you got it wrong. The variable values need single quotes to show they are character values. The quotes do not appear in the printed column or final printing. Also do not get confused between values and the names of variables.

10.

```
int y = 7;          // L1
int x = 8;          // L2
```

```
x = y;              // L3
y = x;              // L4
```

```
System.out.println(x);    // L5
System.out.println(y);    // L6
```

Complete the dry run table (you may not need all rows):

LINE	y	x	printed
L1	7		
L2		8	
L3		7	
L4	7		
L5			7
L6			7

Write EXACTLY what is printed here

7 7

11.

```
String s1 = "rats";           // L1
String s2 = "and";           // L2
String s3 = "cats";          // L3
s1 = s1 + s2;                 // L4
s2 = s2 + s3;                 // L5
s2 = s3;                      // L6
System.out.println(s3);       // L7
```

Complete the dry run table (you may not need all rows):

LINE	s1	s2	s3	printed
L1	"rats"			
L2		"and"		
L3			"cats"	
L4	"ratsand"			
L5		"andcats"		
L6		"cats"		
L7				cats

Write EXACTLY what is printed here

cats

12.

```
String s1 = "cats";           // L1
String s2 = "eat";            // L2
String s3 = "mice";           // L3
s1 = s1 + s2;                  // L4
s2 = s2 + s3;                  // L5
s3 = s2;                       // L6
System.out.println(s3);        // L7
```

Complete the dry run table (you may not need all rows):

LINE	s1	s2	s3	printed
L1	"cats"			
L2		"eat"		
L3			"mice"	
L4	"catseat"			
L5		"eatmice"		
L6			"eatmice"	
L7				eatmice

Write EXACTLY what is printed here

eatmice